v0.2.0-alpha.1 English User Guide

Profiles, lower-screen modules, keyboard and mouse control, and the complete Macro editor

Release and download: https://github.com/mastercook777/Heimdall-AYN-Thor-Assistant/releases/tag/v0.2.0-alpha.1

Contents

Before you begin: Win is a known limitation

Android streaming clients do not reliably forward external Win input. Keypad does not offer Win, and Full Keyboard shows disabled Win keys that emit no output. Use the streaming client's built-in keyboard when Win is required.

Start here: Heimdall still has only three core concepts

If you remember this page, the rest of the app becomes much easier to understand.

Concept	Plain-English meaning	What it owns
Profile	One game / emulator / play scenario	Layout, module content, touch settings, maps, guides, assets, App binding
Home Grid	That Profile's lower-screen desktop	A 6 × 8 editable area: module position and size
Module	A functional building block on the Grid	Macros, Touchpad, Keyboard Pad, Quick Actions, Magnifier, Canvas

In one sentence
Profile = "this game"; Grid = "this game's lower-screen desktop"; Module = "the tool you place on that desktop."

Guide contents

1. What is new in 0.2.0?

New feature	What it is for	Shizuku?
Virtual Mouse	PC streaming, PC emulators, mouse/keyboard games: relative movement, clicks, drag, wheel	Yes
Keyboard Pad	Put 1-12 commonly used PC keys or chords on the lower screen	Yes
Full PC Virtual Keyboard	Temporary full US ANSI keyboard plus navigation / Numpad layer	Yes
Configurable Quick Actions	Up to 3 ordered buttons + optional media-volume slider; can open the virtual keyboard	Depends on action
Controller Macro Composer	Compose D-pad + digital-button motions, charge holds, mirroring, and content copy	Yes

Existing 0.1.1 Profiles still make sense
Legacy Profiles and legacy Quick Actions keep their old defaults. The new features are additive: nothing enters your layout until you add or configure it.

2. Understand the Heimdall main screen

A 0.2.0 real-device layout: map reference, skill Macros, Canvas, and Quick Actions arranged around one game Profile.

☐ Top: current Profile and system status

This area tells you which configuration is active and shows the Profile icon, time, battery, and network status. The Profile area provides quick switching; full management stays in Settings.

☐ Middle: the editable Grid

Macros, Touchpad, Keyboard Pad, Quick Actions, Magnifier, and Canvas can all live here. Move and resize them directly in the Grid editor; you never need to type x/y/w/h coordinates.

☐ Bottom: fixed navigation

Destination	Purpose
Home	Your play-time control console
Map	Local maps / PDFs / Interactive Map for the active Profile
Guide	TXT / Markdown guides and notes
Settings	Profiles, Grid, input, modules, connection, appearance, backup

3. Build your first Profile from scratch

1. Open Settings → Profile management and create a Profile for one game, emulator, or streaming scenario.

2. Optional but recommended: run the target App on the upper screen and bind it with the recent-App picker. App-aware switching changes Profiles only when the match is unambiguous.

3. Choose a starter preset: Balanced, Controls, or Macros. Balanced is the safest first choice.

4. Open the visual Grid editor. Drag modules directly and resize from the visible handle; items snap to the 6 × 8 Grid.

5. Return to Settings and press Save at the bottom. Grid and several Profile settings use a preview-first, explicit-save model.

> The most common mistake
> Seeing the Home screen change does not mean the Grid has been permanently stored. Before leaving the draft workflow or switching Profiles, make sure you pressed Save in Settings.

Which preset should I start with?

Preset	Best for
Balanced	A bit of room for controls, macros, and helpers; recommended for first-time setup
Controls	Large Touchpad / Precision Aim / Virtual Mouse surface
Macros	Emulators, RPGs, fighting games, or any layout with many shortcuts

4. Module overview

Module	What it does	Save boundary
Macro	Upper-screen touch, waits, physical-controller sequences	Macro editor Save; Grid placement separately
Touchpad	Compatible Touch / Enhanced Touch / Right Stick / Precision Aim / Virtual Mouse	Mode and tuning are Profile-owned
Keyboard Pad	PC keys/chords/held keys; can also open Full Keyboard	Component Save; Grid placement via Grid Save
Quick Actions	Screenshot, recording, Full Keyboard, media volume	Component Save; Grid placement via Grid Save
Magnifier	Live upper-screen region on the lower screen	Region/display settings stored with Profile
Canvas	User-supplied static reference image	Composition confirmed separately; Grid placement separately

> Do not try to fit everything
> Heimdall is useful because you can remove what a game does not need. A PC streaming Profile may want Virtual Mouse + Keyboard Pad; an FPS may want a huge Precision Aim surface; an emulator may be better with Macros + Canvas.

4.1 Macro module: turn actions into lower-screen buttons

Macro buttons and Canvas can work side by side: buttons execute, the image provides reference.

Macro steps can include

- Tap a captured upper-screen coordinate
- Hold an upper-screen coordinate
- Swipe between two upper-screen points
- Wait for a duration
- Record and replay real Thor controller input (Shizuku)
- Compose structured D-pad + digital-button controller actions (Shizuku)

Editing

Normal tap = run the Macro. Hold for about 1.8 seconds = open its editor. A Macro can have a label, built-in or imported icon, display emphasis, structured steps, and optional icon-only presentation for compact cells.

Advanced 0.2.0 Macro tools

- Controller compositions support action interval, per-action Hold for charge moves, and final hold timing.

- Mirror flips only left/right D-pad direction while preserving vertical direction, buttons, order, and timing.
- Copy Macro clones content into another existing Macro button without moving the destination Grid slot.
- Only one Macro runs at a time; ordinary repeat taps are not queued. A composed controller Macro with an explicit long Hold can be cancelled by tapping the same Macro again.

> Recording is still the primary route
> The composer is for deterministic digital D-pad/button motions. For analog sticks, analog triggers, and arbitrary physical behavior, record the real controller sequence instead.

4.2 Touchpad: five different input semantics

Precision Aim in real play: the physical right stick handles large turns while the lower screen handles fine correction.

Mode	Actual output	Typical use	Permission
Compatible Touch	Android Accessibility upper-screen drag	Touch-first mobile games / compatibility	Accessibility
Enhanced Touch	Shizuku route designed to coexist with Thor mapping when available	Mapped mobile games	Shizuku

Mode	Actual output	Typical use	Permission
Virtual Right Stick	Thor right-stick axis output	Large touch area controlling camera stick	Shizuku
Precision Aim	Short right-stick correction pulses	ADS, hold angles, head-line correction	Shizuku
Virtual Mouse	Real relative mouse movement/buttons/wheel	PC streaming and mouse/keyboard games	Shizuku

First tuning pass
Use defaults until you know the route works. Then change one value at a time: sensitivity, inversion, curve, wheel distance, or pulse timing.

4.3 Virtual Mouse: a real PC-style touchpad

Virtual Mouse is not Precision Aim. Precision Aim outputs controller-stick corrections; Virtual Mouse creates a separate relative mouse device for Steam Link, Moonlight, PC emulators, and mouse/keyboard-only games.

Standard presentation

- Upper area: one-finger relative pointer movement.
- Integrated lower left/right buttons: press like a laptop touchpad; hold a button and move with another finger to drag.
- Two-finger vertical motion: wheel scrolling.
- One-finger tap in movement area: left click.
- Double tap: two ordinary clicks.
- Second tap + drag: double-tap-drag.
- Two-finger tap without scrolling: right click.

Full gesture area

When enabled, the visible lower buttons and their dedicated hit regions disappear and the same gesture system spans the whole module. It is off by default, so legacy Profiles keep the integrated-button layout.

Useful settings

- Pointer sensitivity
- Invert Y
- Wheel step distance

Client behavior can vary
Virtual Mouse depends on the upper client retaining input focus. Mouse capture and wheel handling can differ between streaming clients, PC games, and emulators.

4.4 Keyboard Pad: hardware-style PC keys on the Grid

Keyboard Pad is the compact, configurable PC-keyboard module. It is not an Android IME and it is not the full keyboard. It is best for keeping a small set of frequent controls—Esc, WASD, function keys, or emulator shortcuts—within easy thumb reach.

You can configure	Details
Key count	1-12; new pads start as a compact F1-F4 four-key block
Packing	Horizontal-first or true vertical 1 × N
Binding	Actual Linux/PC key plus Ctrl, Shift, or Alt modifiers. Win is intentionally unavailable.
Behavior	Press (one down/up) or While held (down until release)
Display	Custom label and built-in icon are separate from the emitted binding
Heimdall Action	A key can open the Full Virtual Keyboard without emitting a Linux key code

How do I edit it?

Keycaps are reserved for keyboard input and never use long-press editing. Hold the top “•••” chassis area for about 1.8 seconds to open the Keyboard Pad editor.

> Keyboard Pad has two different Save boundaries
> Save key content directly in the Keyboard Pad editor. Grid position and module size are still Grid data and require the normal Settings Grid Save.

4.5 Quick Actions: now configurable per module

Freya White on real hardware: Quick Actions can keep capture and volume controls while also exposing the Full PC Virtual Keyboard.

In 0.2.0, each Quick Actions module can hold up to three ordered button actions plus an independently optional media-volume slider. Open Virtual Keyboard can also be assigned as one of those action slots.

Action	What it does
Screenshot	Captures the Thor upper screen
Screen Recording	Records upper-screen video; uses internal playback capture when the game allows it
Open Virtual Keyboard	Opens the temporary full PC keyboard
Media Volume	Optional slider; does not consume one of the three button slots

Editing

Hold the Quick Actions module for about 1.8 seconds. Choose the three slots and the volume-slider state, then Save directly in the component editor.

> Legacy default is preserved
> Old Quick Actions data still resolves to Screenshot + Recording + Media Volume until you explicitly customize it.

4.6 Full PC Virtual Keyboard

The Full Keyboard is a temporary immersive lower-screen layer. It covers the body and Dock below the fixed Header, then returns you to the exact prior Heimdall page when closed. It does not modify Profile or Grid state.

Area / behavior	Details
ABC layer	US ANSI: Esc, F1-F12, number and punctuation rows, QWERTY, Backspace, Tab, Enter, Ctrl, Shift, Alt, and Space. Win is dimmed and emits no output.
Nav / Numpad layer	Large direction cluster, Insert/Delete/Home/End/Page Up/Page Down, true PC Numpad
Hold and repeat	Real key down/hold/up; host owns repeat
Multi-touch chords	Multiple simultaneous keys
Modifiers	Tap to latch immediately; tap again within the short lock window to lock; clear visual state
Release All	Immediately releases all held keyboard state

How to open it

- Put Open Virtual Keyboard in Quick Actions.
- Or assign a Keyboard Pad key to Heimdall Action → Open Virtual Keyboard.

“•••” transport lamp

Color	Meaning
Gray	Connecting / preparing
Green	Keyboard transport is ready
Red	Unavailable; keycaps remain disabled instead of pretending to send input

Not an Android text keyboard
V1 is a fixed PC US ANSI play-time keyboard. It does not replace Gboard and does not participate in Android IME/text-composition lifecycle.

4.7 Magnifier: bring a live upper-screen region down

Circular Magnifier is especially useful for minimaps and HUD elements.

1. Add Magnifier to the Grid and save the layout.
2. Tap the module; Android will ask for screen-capture consent.
3. Draw the desired region directly on the upper screen and confirm it.
4. Hold the module later to reselect or refine the source region.

- Rectangle and true circular presentation.
- Profile-owned zoom and refresh-rate choices.
- Rectangular presentation supports freeze/resume interaction; circular mode keeps its own stop routes.
- One live Magnifier per Profile.
- Magnifier and screen recording currently cannot share MediaProjection at the same time.

4.8 Canvas: pin a static reference image to Home

Canvas is ideal for move lists, control references, boss mechanics, and other static visual notes.

Gesture	Result
Single tap	Brief visual feedback only
Double tap	Temporary full-screen view
Hold ~1.8 s	Replace image or edit composition

- JPG / PNG / static WebP
- Zoom and focal composition
- Rectangle / circular crop
- Multiple independent Canvases per Profile
- 0.2.0 improves runtime sharpness for heavily zoomed Canvas images while keeping bounded decoding

5. Maps and Guides: reference pages outside the Grid

Guides remain on the lower screen while the game stays visible above.

Maps

- Local map images
- PDF maps
- Interactive Map web link
- Multiple local maps in a Profile gallery
- Zoom/pan viewer plus external-open fallback for web content

Guides

- Local TXT / Markdown
- Long-text reading
- Remembered reading position
- Bookmarks
- Good for walkthroughs, bosses, builds, and personal notes

> Canvas vs Map vs Guide
> One image you want visible beside controls → Canvas. A map you want to enter and zoom around → Map. Long text with reading position and bookmarks → Guide.

6. Which permissions do I actually need?

You do not need to enable everything on first launch, and Heimdall does not require root. Enable only the route needed by the features you use.

Feature	Required setup
Profiles / Grid / Canvas / Map / Guide	None beyond installing Heimdall
Basic upper-screen tap/hold/swipe / Compatible Touch	Heimdall Accessibility service
Enhanced Touch / controller macros / Virtual Right Stick / Precision Aim	Shizuku
Virtual Mouse / Keyboard Pad / Full PC Keyboard	Shizuku
Magnifier / screen recording	Android screen-capture consent (MediaProjection)

> Developer options are only for the Shizuku path
> Profiles, Grid, Canvas, Map, Guide, and even Accessibility-based touch do not require Developer options. Wireless debugging is needed to start Shizuku.

6.1 Enable Developer options and wireless debugging

1. Open Thor Android Settings.
2. Open About device / About handheld (names vary), or search for Build number.
3. Tap Build number seven times and enter your lock PIN if Android asks.
4. Return to Settings and open/search Developer options.
5. Enable USB debugging.
6. Open Wireless debugging and turn it on.

6.2 Pair and start Shizuku

1. Install and open the official Shizuku app.
2. Choose Start via wireless debugging and begin pairing.
3. Return to Android Settings → Developer options → Wireless debugging and choose Pair device with pairing code.
4. Enter the pairing code in Shizuku.
5. Return to Shizuku, press Start, and wait until it clearly reports that it is running.

6.3 Authorize Heimdall

1. Open Heimdall → Settings → Connection.

2. Find the Shizuku / controller-enhancement readiness row.

3. Press authorize/connect and allow Heimdall in the Shizuku permission dialog.

4. Return to Heimdall and confirm the Shizuku-powered capability is available. Test one simple action before tuning advanced parameters.

> After a Thor reboot
> Open Shizuku first. If it is no longer running, press Start again; the wireless-debugging pairing normally does not need to be repeated.

7. Advanced physical-controller macros

Use real recording for general controller behavior. Use the structured composer when you need deterministic D-pad motions, digital-button chords, or charge timing that is awkward to perform inside a short recording window.

English structured controller composer: arrange directions, physical buttons, Hold timing, and the action sequence on one surface, then Test before replacing the Macro step.

Tool	When to use it
Record physical controller	Most faithful route; captures real event sequences
Compose controller input	Digital D-pad + digital buttons advertised by the current controller mode
Action interval	Continuous transition timing between actions
Per-action Hold	Keep one direction/button/chord held for 8-5000 ms
Mirror	Flip horizontal D-pad direction only
Copy Macro	Clone label, icon, emphasis, and steps into another existing Macro button
Cancel active hold	Tap the same composed Hold Macro again to interrupt and release inputs

Wait is not Hold
Macro-level Wait = all inputs released. Controller action interval = timing between continuous frames. Per-action Hold = the selected direction/buttons stay physically held.

8. Four practical layout patterns

Action-RPG example: minimap, frequent actions, and reference information share the lower screen.

A. Action RPG

- Circular Magnifier for minimap/HUD
- A few high-frequency Macros
- Touchpad only if the game benefits from touch/camera control
- Quick Actions for capture and volume

B. FPS / TPS

- Give Precision Aim the largest useful area
- Keep physical right stick for large turns
- Only a few critical Macros near the thumb
- Add Magnifier only when a HUD element truly deserves constant attention

C. PC streaming / mouse-and-keyboard game

- Virtual Mouse as the primary surface
- Keyboard Pad with 3-8 high-frequency keys/chords
- Quick Actions slot for Open Virtual Keyboard
- Call Full Keyboard only when complete typing/input is needed

D. Emulator / retro game

- Macros for save/load/fast-forward/emulator actions
- Keyboard Pad for F-keys, number keys, or PC-emulator shortcuts
- Canvas for move lists or control charts
- Map / Guide for full reference material

9. Save boundaries: what saves where?

What you changed	How it is saved
Grid module position / size / add / delete	Preview draft first; final Settings Save
Touchpad mode and tuning	Settings draft; final Settings Save
Macro content	Macro editor Save
Keyboard Pad key content / key count / packing	Keyboard Pad editor Save; no extra Grid Save
Quick Actions slots / volume switch	Quick Actions editor Save; no extra Grid Save
Keyboard Pad / Quick Actions position and size	Still Grid data; requires Settings Grid Save
Canvas composition	Confirm in Canvas editor; Grid position remains separate

Why it is split this way
Module content and Grid placement are different data. 0.2.0 specifically fixes the Keyboard Pad / Quick Actions component boundary so saving component content does not force you to commit an unchanged Grid.

10. Back up, update, and share Profiles

Once you have tuned layouts, Macros, keys, and assets, a Profile is valuable authored data. Export important Profiles after major setup work.

A .heimdall-profile bundle can include

- Profile configuration
- Profile artwork
- Map resources
- Local-file Guides
- Imported Macro icons
- Canvas images
- Structured configuration for new 0.2.0 modules such as Virtual Mouse, Keyboard Pad, and Quick Actions

When updating Heimdall

1. Export important Profiles first.
2. Normal public Alpha updates with the same signing identity can usually install over the existing app.
3. Do not uninstall just to update; uninstalling deletes local Heimdall data.
4. If an advanced input feature is unavailable after an update/reboot, check that Shizuku is running and authorized.

Recovery snapshots are not an external backup
Heimdall keeps a limited set of recovery snapshots for some destructive Profile operations, but those do not replace an exported bundle stored outside the app.

11. Troubleshooting and current limitations

Symptom	Check this first
Grid reverted after restart	Did you press Save at the bottom of Settings?
Keyboard Pad / Quick Actions change disappeared	Did you press Save in the component editor? A genuinely modified Grid draft intentionally blocks ambiguous component saves.
Virtual Mouse / keyboard does nothing	Is Shizuku running and authorized? Does the upper streaming/emulator window still have input focus?
Some PC game ignores Virtual Mouse tap click	0.2.0 uses a compatibility click pulse; individual games may still enforce different minimum click/capture behavior.

Symptom	Check this first
Magnifier and recording cannot run together	They currently share MediaProjection.
Recording has no game audio	Android Audio Playback Capture obeys the game’s own capture policy.
Recorder cannot see a mapped controller button	Thor’s built-in mapper can hide already-mapped physical events from Heimdall.
App-aware did not switch Profiles	Heimdall intentionally does nothing when identity is ambiguous; universal ROM detection is not available.
Win produces no output while streaming	This is a known boundary. Keypad does not offer Win, and Full Keyboard shows disabled Win keys that emit no output. Use the streaming client's built-in keyboard when Win is required.
Full Virtual Keyboard cannot type Chinese Android text	Correct: it is a PC US ANSI play-time keyboard, not an Android IME.

Shizuku troubleshooting
After reboot, open Shizuku first. If it is stuck “searching for pairing service,” verify wireless debugging, background execution, and notifications; toggling wireless debugging off/on can help before reauthorizing Heimdall.

One final question

“What would make this game easier to play if it lived on Thor’s lower screen?”

Heimdall is not about putting every feature on screen. It is about keeping the right extra controls and references within thumb reach.

Good starting points
Mobile FPS: Precision Aim. PC streaming: Virtual Mouse + Keyboard Pad. Emulator: Macro + Canvas. Full PC input: Quick Actions → Full Virtual Keyboard.